# Shuo Guo

The Chinese University of Hong Kong, Shenzhen
School of Science and Engineering
E-Mail: shuoguo@link.cuhk.edu.cn
Personal Web: sshuoguo.github.io


## Education

**The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)** Shenzhen, China
B.Sc. in Applied Mathematics Sept 2025 – May 2029 (expected)
Advised by Prof. Dr. Zhi-Quan Tom Luo

## Research Interests

- Theoretical optimization and the mathematical foundations of machine learning.
- Dynamics, stability, and convergence of stochastic and adaptive optimization algorithms, especially Adam and Muon, with particular interest in dynamical-systems and stochastic-analysis approaches.

## Experience

**Research Assistant, CUHK-Shenzhen** May 2026 – Jul 2026
Benchmarked modern optimization methods for neural-network training, including small-scale language-model experiments and hyperparameter studies.

## Honors and Awards

- Muse Full-tuition Scholarship, CUHK-Shenzhen

## Selected Mathematical Background

- **Convex Optimization:** Studied selected topics from *Lectures on Convex Optimization* by Yurii Nesterov, including convexity, gradient methods, strong convexity, complexity bounds, and accelerated first-order methods.
- **Data Science:** Studied selected topics from *Foundations of Data Science*, including SVD, random graphs, concentration phenomena, and VC dimension, through an independent course led by Prof. Zhiquan Tom Luo.
- **Deep Learning and Optimization:** Studied training dynamics and the theoretical analysis of adaptive optimization methods, with particular emphasis on the Adam family of algorithms; read selected chapters of *Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory*.
- **Stochastic Analysis:** Self-studied measure-theoretic probability and stochastic calculus, including Brownian motion, stochastic integration, stochastic differential equations, and Itô's formula.

## Technical Skills

- **Programming:** Python; numerical and machine-learning experiments.
- **Typesetting:** LaTeX.

## Selected Coursework

- Probability I (Honors); Linear Algebra I and II (Honors); Calculus I and II (Honors); Mathematical Analysis; Ordinary Differential Equations (Honors); Stochastic analysis and Stochastic Differential Equations; Morden Optimization.